

T3 BSM UV terms grouped by field content

EFT order: 5; loop order: 1. Columns are determined only from BSM UV terms. Empty cells denote no UV term with that field combination.

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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	F	S_1	S_2	$F D$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{1}{2}M_F N_{r_1}^T C N_{r_1}$	$+\frac{1}{2}\lambda_{S_1}\text{NewScalar}1_{a_1}\text{NewScalar}1_{a_2}\text{NewScalar}1_{a_1}^\dagger\text{NewScalar}1_{a_2}^\dagger$ $+M_{S_1}^2(-\text{NewScalar}1_{a_1})\text{NewScalar}1_{a_1}^\dagger$	$+\frac{1}{2}\lambda_{S_2}\text{NewScalar}2_{a_1}\text{NewScalar}2_{a_2}\text{NewScalar}2_{a_1}^\dagger\text{NewScalar}2_{a_2}^\dagger$ $-M_{S_2}^2\text{NewScalar}2_{a_1}\text{NewScalar}2_{a_1}^\dagger$	$+\frac{1}{2}iN_{r_1}^T C \gamma^{\mu 1} D_{\mu 1}(N_{r_1})$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{1}{2}M_F N_{A_1, r_1}^T C N_{A_1, r_1}$	$+\frac{1}{2}\lambda_{S_1}\text{NewScalar}1_{a_1}\text{NewScalar}1_{a_2}\text{NewScalar}1_{a_1}^\dagger\text{NewScalar}1_{a_2}^\dagger$ $+M_{S_1}^2(-\text{NewScalar}1_{a_1})\text{NewScalar}1_{a_1}^\dagger$	$+\frac{1}{2}\lambda_{S_2}\text{NewScalar}2_{a_1}\text{NewScalar}2_{a_2}\text{NewScalar}2_{a_1}^\dagger\text{NewScalar}2_{a_2}^\dagger$ $-M_{S_2}^2\text{NewScalar}2_{a_1}\text{NewScalar}2_{a_1}^\dagger$	$+\frac{1}{2}iN_{A_1, r_1}^T C \gamma^{\mu 1} D_{\mu 1}(N_{A_1, r_1})$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-M_F \overline{N_{a_1, r_1}} N_{a_1, r_1}$	$-\text{NewScalar}1 M_{S_1}^2 \text{NewScalar}1^\dagger$ $+\frac{1}{2}\text{NewScalar}1^2 \lambda_{S_1} (\text{NewScalar}1^\dagger)^2$	$+\frac{1}{2}\lambda_{S_2}\text{NewScalar}2_{A_1}\text{NewScalar}2_{A_2}\text{NewScalar}2_{A_1}^\dagger\text{NewScalar}2_{A_2}^\dagger$ $-M_{S_2}^2\text{NewScalar}2_{A_1}\text{NewScalar}2_{A_1}^\dagger$	$+i\overline{N_{a_1, r_1}} \gamma^{\mu 1} D_{\mu 1}(N_{a_1, r_1})$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-M_F \overline{N_{a_1, r_1}} N_{a_1, r_1}$	$+\frac{1}{2}\lambda_{S_1}\text{NewScalar}1_{A_1}\text{NewScalar}1_{A_2}\text{NewScalar}1_{A_1}^\dagger\text{NewScalar}1_{A_2}^\dagger$ $+M_{S_1}^2(-\text{NewScalar}1_{A_1})\text{NewScalar}1_{A_1}^\dagger$	$-\text{NewScalar}2 M_{S_2}^2 \text{NewScalar}2^\dagger$ $+\frac{1}{2}\text{NewScalar}2^2 \lambda_{S_2} (\text{NewScalar}2^\dagger)^2$	$+i\overline{N_{a_1, r_1}} \gamma^{\mu 1} D_{\mu 1}(N_{a_1, r_1})$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\overline{N_{a_1, r_1}} N_{a_1, r_1} M_F$	$-\text{NewScalar}1_{A_1}\text{NewScalar}1_{A_1}^\dagger M_{S_1}^2$ $+\frac{1}{2}\text{NewScalar}1_{A_1}\text{NewScalar}1_{A_2}\lambda_{S_1}\text{NewScalar}1_{A_1}^\dagger\text{NewScalar}1_{A_2}^\dagger$	$-M_{S_2}^2\text{NewScalar}2_{A_1}\text{NewScalar}2_{A_1}^\dagger$ $+\frac{1}{2}\text{NewScalar}2_{A_1}\text{NewScalar}2_{A_2}\lambda_{S_2}\text{NewScalar}2_{A_1}^\dagger\text{NewScalar}2_{A_2}^\dagger$	$+i\overline{N_{a_1, r_1}} \gamma^{\mu 1} D_{\mu 1}(N_{a_1, r_1})$

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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 D$	$S_1 H$	$S_1 S_2$	$S_2 D$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+D_{\mu_1}(\text{NewScalar1}_{a_1})D_{\mu_1}(\text{NewScalar1}_{a_1})^\dagger + \lambda_{H1}H_{a_1}\text{NewScalar1}_{a_2}H_{a_1}^\dagger\text{NewScalar1}_{a_2}^\dagger + \lambda_{12}\text{NewScalar1}_{a_1}\text{NewScalar2}_{a_2}\text{NewScalar1}_{a_1}^\dagger\text{NewScalar2}_{a_2}^\dagger + D_{\mu_1}(\text{NewScalar2}_{a_1})D_{\mu_1}(\text{NewScalar2}_{a_1})^\dagger$			
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+D_{\mu_1}(\text{NewScalar1}_{a_1})D_{\mu_1}(\text{NewScalar1}_{a_1})^\dagger + \lambda_{H1}H_{a_1}\text{NewScalar1}_{a_2}H_{a_1}^\dagger\text{NewScalar1}_{a_2}^\dagger + \lambda_{12}\text{NewScalar1}_{a_1}\text{NewScalar2}_{a_2}\text{NewScalar1}_{a_1}^\dagger\text{NewScalar2}_{a_2}^\dagger + D_{\mu_1}(\text{NewScalar2}_{a_1})D_{\mu_1}(\text{NewScalar2}_{a_1})^\dagger$			
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+D_{\mu_1}(\text{NewScalar1})D_{\mu_1}(\text{NewScalar1})^\dagger + \text{NewScalar1}\lambda_{H1}\text{NewScalar1}^\dagger H_{a_1}H_{a_1}^\dagger + \lambda_{12}\text{NewScalar1}\text{NewScalar1}^\dagger\text{NewScalar2}_{A_1}\text{NewScalar2}_{A_1}^\dagger + D_{\mu_1}(\text{NewScalar2}_{A_1})D_{\mu_1}(\text{NewScalar2}_{A_1})^\dagger$			
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+D_{\mu_1}(\text{NewScalar1}_{A_1})D_{\mu_1}(\text{NewScalar1}_{A_1})^\dagger + \lambda_{H1}H_{a_1}\text{NewScalar1}_{A_1}H_{a_1}^\dagger\text{NewScalar1}_{A_1}^\dagger + \lambda_{12}\text{NewScalar2}\text{NewScalar2}^\dagger\text{NewScalar1}_{A_1}\text{NewScalar1}_{A_1}^\dagger + D_{\mu_1}(\text{NewScalar2})D_{\mu_1}(\text{NewScalar2})^\dagger$			
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+D_{\mu_1}(\text{NewScalar1}_{A_1})^\dagger D_{\mu_1}(\text{NewScalar1}_{A_1}) + H_{a_1}\text{NewScalar1}_{A_1}\lambda_{H1}H_{a_1}^\dagger\text{NewScalar1}_{A_1}^\dagger + \text{NewScalar1}_{A_1}\text{NewScalar2}_{A_2}\lambda_{12}\text{NewScalar1}_{A_1}^\dagger\text{NewScalar2}_{A_2}^\dagger + D_{\mu_1}(\text{NewScalar2}_{A_1})^\dagger D_{\mu_1}(\text{NewScalar2}_{A_1})$			

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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_2 H$	$S_1 F \ell$	$S_1 S_2 H$	$S_2 F \ell$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\lambda_{H2} H_{a_1} \text{NewScalar2}_{a_2} H_{a_1}^\dagger \text{NewScalar2}_{a_2}^\dagger$	$+\text{NewScalar1}_{a_1}^\dagger \overline{y_1, i_1, r_1} N_{r_1}^T C P_L \ell_{a_1, i_1}$ $+\text{NewScalar1}_{a_1} y_1, i_1, r_1 \overline{\ell_{a_1, i_1}} P_R N_{r_1}$	$+\lambda_{T3} \text{NewScalar2}_{a_1} \epsilon_{a_2, a_3} H_{a_1}^\dagger H_{a_2}^\dagger \text{NewScalar1}_{a_3}^\dagger$ $+\lambda_{T3} H_{a_1} H_{a_2} \text{NewScalar1}_{a_3} \text{NewScalar2}_{a_1}^\dagger \bar{\epsilon}_{a_2, a_3}$	$\text{NewScalar2}_{a_2} \bar{\epsilon}_{a_1, a_2} \overline{y_2, i_1, r_1} N_{r_1}^T C P_L \ell_{a_1, i_1}$ $+\epsilon_{a_2, a_1} y_2, i_1, r_1 \text{NewScalar2}_{a_1}^\dagger \overline{\ell_{a_2, i_1}} P_R N_{r_1}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\lambda_{H2} H_{a_1} \text{NewScalar2}_{a_2} H_{a_1}^\dagger \text{NewScalar2}_{a_2}^\dagger$	$+\text{NewScalar1}_{a_1}^\dagger \overline{y_1, i_1, r_1} \text{gen}(\text{SU2L}(\text{fund}))_{A_1, a_1, a_2} N_{A_1, r_1}^T C P_L \ell_{a_2, i_1}$ $+\text{NewScalar1}_{a_2} y_1, i_1, r_1 \text{gen}(\text{SU2L}(\text{fund}))_{A_1, a_1, a_2} \overline{\ell_{a_1, i_1}} P_R N_{A_1, r_1}$	$+\lambda_{T3} \text{NewScalar2}_{a_1} \epsilon_{a_2, a_3} H_{a_1}^\dagger H_{a_2}^\dagger \text{NewScalar1}_{a_3}^\dagger$ $+\lambda_{T3} H_{a_1} H_{a_2} \text{NewScalar1}_{a_3} \text{NewScalar2}_{a_1}^\dagger \bar{\epsilon}_{a_2, a_3}$	$\text{NewScalar2}_{a_2} \bar{\epsilon}_{a_1, a_2} \overline{y_2, i_1, r_1} \text{gen}(\text{SU2L}(\text{fund}))_{A_1, a_1, a_3} N_{A_1, r_1}^T C P_L \ell_{a_3, i_1}$ $+\epsilon_{a_2, a_1} y_2, i_1, r_1 \text{NewScalar2}_{a_1}^\dagger \text{gen}(\text{SU2L}(\text{fund}))_{A_1, a_3, a_2} \overline{\ell_{a_3, i_1}} P_R N_{A_1, r_1}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\lambda_{H2} H_{a_1} \text{NewScalar2}_{A_1} H_{a_1}^\dagger \text{NewScalar2}_{A_1}^\dagger$	$+\text{NewScalar1}^\dagger \bar{\epsilon}_{a_1, a_2} \overline{y_1, i_1, r_1} \ell_{a_1, i_1}^T C P_L N_{a_2, r_1}$ $+\text{NewScalar1} \epsilon_{a_1, a_2} y_1, i_1, r_1 \overline{\ell_{a_1, i_1}} C P_R \overline{N_{a_2, r_1}}^T$	$+\lambda_{T3} \text{NewScalar1}^\dagger \text{NewScalar2}_{A_1} \epsilon_{a_3, a_2} H_{a_1}^\dagger H_{a_2}^\dagger \text{gen}(\text{SU2L}(\text{fund}))_{A_1, a_1, a_3}$ $+\text{NewScalar1} \lambda_{T3} H_{a_2} H_{a_3} \text{NewScalar2}_{A_1}^\dagger \bar{\epsilon}_{a_1, a_2} \text{gen}(\text{SU2L}(\text{fund}))_{A_1, a_1, a_3}$	$\text{NewScalar2}_{A_1} \overline{y_2, i_1, r_1} \text{gen}(\text{SU2L}(\text{fund}))_{A_1, a_1, a_2} \overline{N_{a_1, r_1}} P_L \ell_{a_2, i_1}$ $+y_2, i_1, r_1 \text{NewScalar2}_{A_1}^\dagger \text{gen}(\text{SU2L}(\text{fund}))_{A_1, a_1, a_2} \overline{\ell_{a_1, i_1}} P_R N_{a_2, r_1}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+\text{NewScalar2} \lambda_{H2} \text{NewScalar2}^\dagger H_{a_1} H_{a_1}^\dagger$	$\text{NewScalar1}_{A_1}^\dagger \bar{\epsilon}_{a_1, a_2} \overline{y_1, i_1, r_1} \text{gen}(\text{SU2L}(\text{fund}))_{A_1, a_2, a_3} \ell_{a_3, i_1}^T C P_L N_{a_1, r_1}$ $+\text{NewScalar1}_{A_1} \epsilon_{a_1, a_2} y_1, i_1, r_1 \text{gen}(\text{SU2L}(\text{fund}))_{A_1, a_3, a_2} \overline{\ell_{a_3, i_1}} C P_R \overline{N_{a_1, r_1}}^T$	$+\lambda_{T3} \text{NewScalar2}^\dagger \text{NewScalar1}_{A_1} H_{a_2} H_{a_3} \bar{\epsilon}_{a_1, a_2} \text{gen}(\text{SU2L}(\text{fund}))_{A_1, a_1, a_3}$ $+\text{NewScalar2} \lambda_{T3} \epsilon_{a_3, a_2} H_{a_1}^\dagger \text{NewScalar1}_{A_1}^\dagger H_{a_2}^\dagger \text{gen}(\text{SU2L}(\text{fund}))_{A_1, a_1, a_3}$	$+\text{NewScalar2} y_2, i_1, r_1 \overline{N_{a_1, r_1}} P_L \ell_{a_1, i_1}$ $+\text{NewScalar2}^\dagger y_2, i_1, r_1 \overline{\ell_{a_1, i_1}} P_R N_{a_1, r_1}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+H_{a_1} \text{NewScalar2}_{A_1} \lambda_{H2} H_{a_1}^\dagger \text{NewScalar2}_{A_1}^\dagger$	$+\overline{\ell_{a_3, i_1}} C P_R \overline{N_{a_1, r_1}}^T \text{NewScalar1}_{A_1} y_1, i_1, r_1 \epsilon_{a_1, a_2} \text{gen}(\text{SU2L}(\text{fund}))_{A_1, a_3, a_2}$ $+\overline{y_1, i_1, r_1} \ell_{a_3, i_1}^T C P_L N_{a_1, r_1} \text{gen}(\text{SU2L}(\text{fund}))_{A_1, a_2, a_3} \bar{\epsilon}_{a_1, a_2} \text{NewScalar1}_{A_1}^\dagger$	$+\overline{N_{a_2, r_1}} \text{NewScalar2}_{A_1} \epsilon_{a_3, a_2} \text{Strut}(\text{SU2L})_{A_1, A_2, A_3} \text{gen}(\text{SU2L}(\text{fund}))_{A_3, a_1, a_2} H_{a_1}^\dagger H_{a_2}^\dagger \text{NewScalar1}_{A_1}^\dagger$ $-H_{a_2} H_{a_3} \text{NewScalar1}_{A_1} y_1, i_1, r_1 \text{Strut}(\text{SU2L})_{A_1, A_2, A_3} \text{gen}(\text{SU2L}(\text{fund}))_{A_3, a_1, a_2} \bar{\epsilon}_{a_1, a_2} \text{NewScalar2}_{A_1}^\dagger$	$+\overline{y_2, i_1, r_1} \overline{N_{a_1, r_1}} P_L \ell_{a_2, i_1} \text{NewScalar2}_{A_1} \text{gen}(\text{SU2L}(\text{fund}))_{A_1, a_1, a_2}$ $+\overline{\ell_{a_1, i_1}} P_R N_{a_2, r_1} y_2, i_1, r_1 \text{gen}(\text{SU2L}(\text{fund}))_{A_1, a_1, a_2} \text{NewScalar2}_{A_1}^\dagger$